Assignment No: 7

Title: Use Docker Compose to manage multiple containers for a web-based AI application, including containers for the AI model, database, and frontend.

Steps:

Step 1: Create Project Directory Structure

Open your terminal and create the folders to isolate each service.

```
None

mkdir ai_web_app && cd ai_web_app mkdir ai_model
frontend database
```

Step 2: Create AI Model Service

1. **Create ai_model/app.py**

```
None

from flask import Flask, jsonify
import numpy as np app =
Flask(__name__)

@app.route("/") def predict(): result = np.mean([10, 20,
30, 40]) return jsonify({"prediction": result})

if __name__ == "__main__": app.run(host="0.0.0.0",
    port=5000)
```

2. **Create ai_model/requirements.txt**: Add flask and numpy to this file.

```
None None flask

numpy
```

3. **Create ai_model/Docker file**:

```
FROM python:3.10-slim
WORKDIR /app COPY
requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt COPY
app.py .  CMD ["python", "app.py"]
```

Step 3: Create Frontend Service

1. **Create frontend/index.html**:

```
None
<!DOCTYPE html>
<html>
<head><title>AI Web App</title></head>
<body>
 <h1>Web-Based AI Application</h1>
 <p>AI Model Output will be displayed here</p>
</body>  </html>
```

2. **Create frontend/Dockerfile**:

```
None
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

Step 4: Create docker-compose.yml

In the **root directory** (ai_web_app/), create docker-compose.yml:

YAML

```
None
services: ai_model:
```

```
build: ./ai_model ports:
- "5000:5000" database:
image: postgres:15 environment:
POSTGRES_USER: admin
POSTGRES_PASSWORD: admin123
POSTGRES_DB: aidb volumes: -
db_data:/var/lib/postgresql/data frontend:
build: ./frontend ports:
- "8080:80" depends_on:
- ai_model

volumes: db_data:
```

Step 5: Build and Run

Run the following command from the project root:

```
None

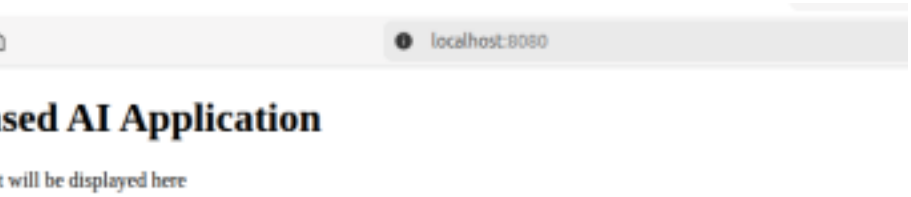
docker compose up --build
```

- **AI Model**: Access http://localhost:5000 to see the JSON prediction. ● **Frontend**: Access http://localhost:8080 to see the webpage

Output:



1. AI Model





2. Frontend

3. Containers running